



Radiological characterisation and inter-laboratory comparison of naturally occurring radioactive materials (NORM) in building materials used in Ireland

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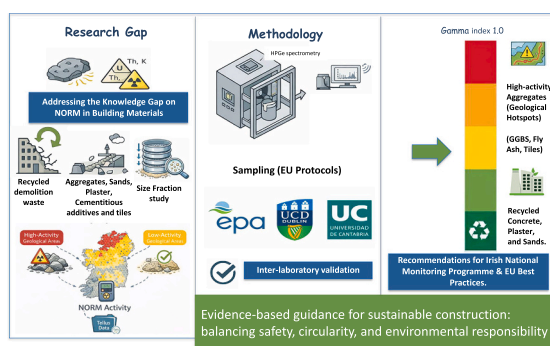
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HIGHLIGHTS

- Natural radioactivity in construction materials was assessed using European screening criteria.
- Gamma spectrometry demonstrated reproducibility suitable for regulatory monitoring.
- Most materials exhibited Gamma Index values ≤ 1 , supporting unrestricted use under EU-BSS.
- Elevated Gamma Index values in one aggregate highlights the need for quarry-specific screening.
- Evidence to support risk-based screening within European circular-economy policy frameworks.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Naturally occurring radioactive materials
NORM
Gamma spectrometry
Building materials
Radiological risk
Gamma Index

ABSTRACT

Naturally occurring radioactive materials (NORM), primarily derived from the uranium-238 and thorium-232 decay series and potassium-40, are present in raw and processed construction materials and can contribute to indoor gamma radiation exposure. This study evaluates the activity concentrations of Ra-226, Th-232 and K-40 in building materials used in Ireland and assesses their radiological suitability for construction applications.

Twenty representative building materials were analysed in two phases. The first phase included aggregates, construction sands, plaster and concretes from pre-1964 residential demolition sites. The second phase focused on twelve materials, including particle-size fractions of concrete and aggregates, and aggregates from low- and high-radioactivity regions, cementitious additives (ground granulated blast-furnace slag (GGBS) and two fly ash samples), and imported ceramic tiles. High-resolution gamma spectrometry measurements were conducted in up to three laboratories (Environmental Protection Agency, EPA; University College Dublin, UCD; and University of

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Cantabria, UC) for inter-laboratory comparison. The Gamma Index (I_γ) was calculated in accordance with European Commission guidelines.

Ra-226 activity concentrations ranged from 5.5 to 169 Bq·kg⁻¹, Th-232 from below detection limits to 206 Bq·kg⁻¹, and K-40 from 5.4 to 1155 Bq·kg⁻¹. Concrete, plaster, sands and low-activity aggregates exhibited low radionuclide levels, with I_γ values ≤ 0.4 , well below the screening limit for unrestricted use. Particle-size fractionation had little effect on radionuclide concentrations in concrete, while greater variation was observed in Ra-226 in aggregate fractions. One aggregate source showed elevated activity ($I_\gamma \approx 2.0 \pm 0.03$), highlighting the importance of geological screening. Cementitious additives and ceramic tiles showed higher activity levels than concrete but remained below regulatory limits.

This study supports more sustainable construction practices by providing data for the safe reuse and recycling of building materials. The findings expand the available NORM dataset for Irish construction materials and support their safe use in the built environment.

1. Introduction

Naturally occurring radioactive materials (NORM), notably radionuclides from the uranium-238 (U-238) and thorium-232 (Th-232) decay series, as well as potassium-40 (K-40), are ubiquitous in geological raw materials (UNSCEAR, 2000; Rodrigues e Silva and Monteiro de Oliveira, 2023). When such materials (e.g., aggregates, sand, cement, concrete, natural stones, recycled building materials) are used in construction, they can contribute to indoor gamma radiation exposure and, in certain cases, a non-negligible contribution to indoor radon concentration, as construction materials rich in Ra-226 can release radon gas into the indoor environment, thus potentially increasing long-term radiological dose to building occupants. Several studies across Europe and beyond have documented this phenomenon, including comprehensive reviews of the radioactivity of building materials and their contribution to indoor exposure (Anagnostakis et al., 2004; Beretka and Mathew, 1985; Trevisi et al., 2012; Papastefanou et al., 2005).

Large-scale surveys across the European Union have documented wide variability in NORM activity concentrations in building materials and associated exposure scenarios (UNSCEAR, 2008), while national studies in Ireland remain limited (Lee et al., 2004). In addition, coordinated European initiatives such as the e-NORM survey have assessed exposure situations involving naturally occurring radioactive materials across multiple EU Member States, highlighting substantial differences in exposure scenarios, regulatory approaches, and the radiological relevance of NORM between countries (Mrdakovic Popic et al., 2023). More recently, a systematic methodology for the identification and prioritisation of NORM has been proposed at the European level, providing a harmonised framework to support regulatory decision-making and risk-based screening approaches (Michalik et al., 2023; Djabou et al., 2026). Complementary databases comprising thousands of samples of bulk and superficial construction materials further demonstrate the wide range of Ra-226, Th-232, and K-40 activity concentrations observed across Europe (Madruca et al., 2019; Trevisi et al., 2012).

To address this gap, the NORM-BMI project was initially undertaken as a baseline assessment of NORM activity in commonly used and recycled building materials. Activity concentrations were measured and compared across three laboratories: The Environmental Protection Agency (EPA, Ireland), University College Dublin (UCD, Ireland), and the University of Cantabria (UC, Spain), providing preliminary insights into radionuclide variability and measurement reproducibility, and highlighting the need for more comprehensive, systematic evaluations in subsequent project phases.

Building on these findings, this study, conducted in two phases, aims to expand the dataset of materials widely used in Irish construction and to evaluate measurement consistency across laboratories. Identical samples were analysed independently by the Environmental Protection Agency (EPA), University College Dublin (UCD), and University of Cantabria (UC) laboratories. Ensuring inter-laboratory comparability is essential for reliable national monitoring procedures and evidence-based regulatory guidance.

Materials selected for the first phase included two types of aggregates

from suppliers, construction sands, and concrete samples collected from residential demolition sites dating back to 1964. These materials were chosen not only because of their widespread use in Irish construction and relevance to long-term indoor exposure scenarios, but also to investigate the potential for safely recycling demolition-derived building materials. This approach allows assessment of whether the reuse of such materials could contribute to sustainability goals by reducing landfill waste, while ensuring that Gamma Index values and radiological safety remain within acceptable limits. In total, eight unique samples were analysed, with each sample measured sequentially by the EPA, UCD, and UC laboratories, resulting in 24 individual measurements.

This study reports activity concentrations of radionuclides from the Ra-226 and Th-232 decay series, as well as K-40, together with the corresponding Gamma Index (I_γ) for each material type. The results provide an expanded understanding of NORM variability in Irish building materials and validate the robustness and reproducibility of inter-laboratory measurements within the NORM-BMI project.

The objectives of the second phase were threefold: (i) to investigate the influence of particle size (grain size fraction) on radionuclide activity concentrations in concrete and aggregate; (ii) to assess the variability of NORM activity associated with high- and low-activity aggregate sources, as identified from national geological (Tellus) data (Geological Survey Ireland, 2014); and (iii) to evaluate cementitious additives, including ground granulated blast furnace slag (GGBS) and fly ash, due to their increasing use in low-carbon construction and their potential to contain elevated NORM levels. Twelve representative samples were analysed, including concrete, aggregates, cementitious additives, and imported ceramic tiles commonly used in Ireland. Unlike the first phase, samples in this phase were measured independently by a single laboratory (EPA, UCD, or UC) using harmonised measurement protocols, reflecting the more targeted scope of this phase.

In addition, the inclusion of materials recovered from residential demolition sites in the first phase provides valuable evidence for assessing the radiological safety of recycled construction materials. As the reuse and recycling of demolition waste are increasingly promoted within European circular-economy and sustainability strategies, understanding the NORM content of such materials is essential for their safe incorporation into new construction. By examining particle-size effects, aggregate source variability, and cementitious additives, the second phase provides essential data to support evidence-based assessment of radiological risk in Irish construction materials. The results also contribute to the safe and sustainable use of building materials, inform potential regulatory guidance, enhance national understanding of NORM distribution, and demonstrate the potential for recycling demolition-derived materials, supporting European initiatives for waste reduction, circular economy, and a greener built environment.

2. Research methodology

2.1. Material selection

The materials were selected to represent the most commonly used

building materials in Ireland, including recycled and demolition-derived products. While the dataset does not cover all possible materials, it provides a robust baseline for radiological screening, with limitations in generalising the results acknowledged due to regional variations in NORM concentrations.

2.1.1. First phase

Materials for this phase were selected to achieve two main objectives: to benchmark the performance of the three participating laboratories (EPA, UCD, UC) on identical samples, and to expand the dataset of NORM concentrations in building materials commonly used in Ireland. Selection criteria combined scientific relevance, representativeness, and logistical feasibility. Priority was given to materials frequently used in the Irish construction sector, including aggregates, construction sands, and concrete types from both demolition and new construction sources. Practical considerations, such as sample availability and representativeness of typical construction practices, also guided the selection process.

The samples for this phase comprised two aggregate samples sourced from major commercial quarries in Ireland, construction sands from two suppliers, and three concrete and plaster samples obtained from residential demolition sites dating back to 1964. Additional concrete samples were included to reflect historical variations in construction practices. In total, eight samples were selected and shipped sequentially to the three laboratories (EPA → UCD → UC) in sealed 500 mL containers to minimise potential loss or contamination and ensure comparability of measurements.

These materials provide a diverse representation of Irish building products, encompassing raw aggregates, finished construction materials, and demolition-derived samples. This selection allowed for a comprehensive assessment of Ra-226, Th-232, and K-40 activity concentrations and the corresponding Gamma Index (I_γ). The resulting dataset supports both validation of laboratory performance and expansion of national NORM screening data to inform regulatory frameworks and public health decision-making.

2.1.2. Second phase

Materials for the second phase were selected to achieve main objectives: to investigate the influence of particle size on radionuclide activity concentrations in concrete; to assess the variability of NORM activity associated with high- and low-activity aggregate sources identified from national geological (Tellus) data; to evaluate cementitious additives, including GGBS and fly ash, due to their increasing use in low-carbon construction and potential for elevated NORM levels; and to assess the contribution of imported construction products commonly used in Ireland.

The sample set for this phase comprised 12 representative materials. Concrete samples were separated into three grain size fractions (<1.18 mm, 1.18–2 mm, and >2 mm). In addition, one aggregate sample was subdivided into two particle size fractions (approximately 1 mm and 1 cm) to assess particle-size effects on radionuclide activity concentrations. Aggregates were sourced from regional quarries with predicted high- and low-NORM activity levels. Cementitious additives included GGBS and two fly ash samples obtained from power stations. Two imported ceramic tiles from Spain and Italy were also included to evaluate the potential contribution of imported construction materials to indoor NORM exposure.

Unlike the first phase, each sample in this phase was measured only once in each participating laboratory (EPA, UCD and UC), reflecting the streamlined scope of the study while still providing robust data for comparative assessment.

2.2. Sample preparation

For all phases of this study, samples were prepared using harmonised procedures to ensure uniformity and comparability across laboratories.

Initially, all samples were manually cleaned to remove surface dust, coatings, and any adhering organic material. The materials were then mechanically reduced to coarse fragments of approximately 3 cm in size using a combination of hammers, chisels, and concrete compression equipment.

The coarse fragments were subsequently ground into fine powder, achieving a particle size of less than 1 mm. This fine homogenisation is in line with IAEA recommendations for gamma spectrometric analysis (IAEA, 2010) and ensures consistent measurement conditions across laboratories. After milling, the powders were sieved to obtain uniform particle size and transferred into sealed containers for shipment.

Upon arrival at the EPA, UCD, and UC laboratories, aliquots from each sample were prepared in counting geometries appropriate for the detector geometry calibration specific to each laboratory. All samples were sealed and stored for a minimum of 28 days prior to measurement to allow secular equilibrium to establish between Ra-226 and its short-lived progeny Pb-214 and Bi-214 via Rn-222 (half-life 3.82 days). This waiting period, widely adopted in gamma spectrometry protocols, ensures that Ra-226 activity can be accurately inferred from its decay products, following established gamma spectrometry protocols and ISO 18589-2 guidelines for environmental radioactivity measurements (ISO, 2015; IAEA, 2010).

2.3. Gamma spectrometry measurements

Gamma spectrometry analyses for first phase were conducted at the three participating laboratories using high-purity germanium (HPGe) detectors. For the second phase, the same protocols were applied, but analyses were carried out only at the UCD and EPA laboratories. Each laboratory employed independent systems and calibration protocols to ensure inter-laboratory comparability.

At the University of Cantabria (UC), measurements were performed using a planar lithium-drifted HPGe detector (GeLi HP, Canberra Model 7229-3020) with a relative efficiency of 20% at the 1332 keV gamma line of Co-60. The detector demonstrated good performance across the energy range, with typical absolute efficiencies of 5% at 59.5 keV (Am-241), 0.9% at 662 keV (Cs-137), and 0.4% at 1460 keV (K-40). Energy resolution was <1%, corresponding to 0.45 keV at 59.5 keV and 1.90 keV at 1332 keV, enabling reliable separation of closely spaced gamma lines.

The detector operated in a low-background environment and was calibrated for both energy and absolute full-energy peak efficiency using certified multi-gamma reference sources covering the energy range of interest (30–3000 keV). Efficiency calibration was performed using the same geometry as the samples (250 mL containers) to ensure consistency in efficiency determination. The calibration was verified periodically to ensure stability and accuracy of the measurements.

Gamma energies from 30 to 3000 keV were recorded, with samples placed in 250 mL geometry containers for 2 h/measurement. Corrections for true coincidence summing and self-attenuation were applied using Genie 2000 software (Canberra Industries, 2000), incorporating known nuclear decay schemes and sample composition data (Quindós et al., 2006). At the EPA Laboratory, supplementary measurements were carried out using a planar HPGe detector (ORTEC GMX-7023-S) with a relative efficiency of 70% at 1.33 MeV (Co-60) and high energy resolution (FWHM of 122 eV at 14.4 keV and 1.22 keV at 1332 keV). Samples were counted in 250 mL cylindrical containers for 24 h following a minimum 28-day storage period to allow secular equilibrium between Ra-226 and its progeny. Efficiency calibration employed certified multi-gamma reference standards. Spectrum acquisition and analysis were conducted with Genie 2000 and Apex-Gamma modules. Corrections for true coincidence summing and self-attenuation were performed using GESPECOR software (Sima et al., 2001).

At University College Dublin (UCD), measurements utilised an HPGe detector (Ortec GMX-15190-SV, relative efficiency 19% at 1.33 MeV) coupled with an Ortec DSPEC 50 digital signal processor. Samples were

measured in 250 mL geometry containers after at least 30 days of storage to ensure secular equilibrium, with counting durations of 24 h/sample to enhance statistical reliability. Calibration used certified multi-gamma reference standards, and corrections for efficiency, true coincidence summing, and self-attenuation were applied using GESPECOR and EFFTRAN software (ORTEC, 2010). Sample-specific self-attenuation corrections were determined via transmission measurements across a range of energies.

Measurement procedures adhered to CEN/TS 17216:2018 (CEN, 2018). The inter-laboratory comparison was evaluated in the context of recent European and international exercises, including NORMCON-STRUCT (Paepen et al., 2024) and the IAEA NORM intercomparison (Karfopoulos et al., 2025), to benchmark laboratory performance and ensure methodological robustness in the assessment of NORM activity concentrations in building materials.

The minimum detectable activity (MDA) for each radionuclide was determined in accordance with Currie's criteria (Currie, 1968), as implemented in the respective analysis software used by each laboratory. Typical MDA ranges ($\text{Bq}\cdot\text{kg}^{-1}$), depending on counting time, detector efficiency, and sample characteristics, are summarised below.

At the University of Cantabria (UC), MDAs ranged from 3.5 to 5.4 for Ra-226, 2.6 to 4.1 for Th-232, and 3.8 to 13.4 for K-40.

At the Environmental Protection Agency (EPA), MDAs were calculated using Apex Gamma software and ranged from 0.71 to 2.54 for Ra-226, 1.06 to 5.14 for Th-232, and 3.53 to 14.7 for K-40.

At University College Dublin (UCD), for a typical counting time of 86,400 s (24 h), MDAs were 1.1–3.1 for Ra-226 and 0.22–0.65 for Th-232 (via Ac-228), while K-40 ranged from 0.9 to 1.3 $\text{Bq}\cdot\text{kg}^{-1}$. Additional progeny radionuclides used for activity determination (e.g. Pb-214, Bi-214, Pb-212) exhibited MDAs below 0.4 $\text{Bq}\cdot\text{kg}^{-1}$.

2.4. Activity determination of Ra-226, Th-232, and K-40

The activity concentration $A(E)$ of radionuclides in the study samples, expressed in Becquerels per kilogram ($\text{Bq}\cdot\text{kg}^{-1}$), was calculated using the equation (Knoll, 2010; Gilmore, 2008):

$$A(E) = \frac{C_{\text{net}}}{\epsilon(E) \times I \times t \times m} \quad (1)$$

where C_{net} is the net count rate of a specific gamma line after background subtraction, $\epsilon(E)$ is the full-energy peak efficiency at energy E , I is the gamma emission probability, t is the measurement time in seconds, and m is the sample mass in kilograms (Djabou et al., 2022; Quindós et al., 2006).

Ra-226 and Th-232 activities were determined indirectly through the gamma emissions of their decay products, assuming secular equilibrium within their respective decay chains. For Ra-226, gamma lines from Pb-214 at 295.1 keV (19.2%) and 351.9 keV (37.1%), as well as from Bi-214 at 1764.5 keV (15.4%), were primarily used at UC to minimise true coincidence summing effects. UCD additionally employed the 609.3 keV gamma line of Bi-214 (45.5%) due to its strong intensity and clear spectral presence. At the EPA, multiple Bi-214 lines (609, 1120, and 1764 keV) were included for Ra-226 assessment.

For Th-232, gamma emissions from Ac-228 at 338.4 keV (12%) and Pb-212 at 238.6 keV (44.6%) were used. At UCD, quantification relied solely on the 338.4 keV line, while EPA measurements included several Ac-228 lines (338.4, 911, and 968 keV). UC used the 338.4 keV line for Th-232 activity, verified by the 911 keV line.

K-40 activity was measured directly from its 1460.8 keV gamma emission, which has a 10.7% emission probability. All full-energy peaks were corrected for potential spectral interferences from overlapping nuclide peaks to ensure accurate results.

2.5. The Gamma Index (I_γ)

The Gamma Index (I_γ) was calculated as a conservative screening tool to assess the radiological suitability of building materials, following European Commission guidance (European Commission, 2014; CEN, 2022). The index provides a preliminary indication of potential indoor exposure, and when $I_\gamma > 1$, a detailed dose assessment is required, accounting for material density, thickness, building type, and intended use bulk or superficial (Markkanen, 1995; European Commission, 1999). The index is defined as:

$$I_\gamma = A_{\text{Ra}}/300 + A_{\text{Th}}/200 + A_{\text{K}}/3000 \quad (2)$$

where A_{Ra} , A_{Th} , and A_{K} are the activity concentrations of Ra-226, Th-232, and K-40 in $\text{Bq}\cdot\text{kg}^{-1}$. The combined standard uncertainty of I_γ was propagated as:

$$u(I_\gamma) = \sqrt{\left[(u(A_{\text{Ra}})/300)^2 + (u(A_{\text{Th}})/200)^2 + (u(A_{\text{K}})/3000)^2 \right]} \quad (3)$$

where $u(A_{\text{Ra}})$, $u(A_{\text{Th}})$, and $u(A_{\text{K}})$ are the standard uncertainties of the respective activity concentrations; uncertainties are reported in Tables 2 and 3. According to the guidelines:

- $I_\gamma \leq 1$: Material is considered safe for use in bulk applications (e.g., concrete). Note that this criterion applies specifically to bulk building materials. For surface materials (e.g., ceramic tiles), a higher reference level of $I_\gamma = 6$ is specified (European Commission, 1999), reflecting their limited contribution to total room exposure.
- $1 < I_\gamma \leq 6$: Restricted use may be needed, and detailed dose assessment is recommended.
- $I_\gamma > 6$: Material is not suitable for construction due to high gamma radiation levels.

While the Gamma Index (I_γ) provides a screening measure for external gamma exposure, it does not account for inhalation of radon (Rn-222) emanating from materials with elevated Ra-226 activity. Radon gas can diffuse from building materials into indoor air and contribute to the overall effective dose, particularly in poorly ventilated spaces. Although the materials analysed in this study generally exhibit low Ra-226 activity, it is noted that higher Ra-226 content can result in increased radon exhalation (Righi and Bruzzi, 2006; El-Taher, 2010; Papaefthymiou and Gouseti, 2008). Therefore, radon exposure should be considered in comprehensive indoor radiological assessments, especially for materials with Ra-226 activity significantly above typical background levels (Turhan, 2008; Fuente et al., 2020).

3. Results and discussion

3.1. First phase

Activity concentrations of Ra-226, Th-232 and K-40 in all first phase samples were measured using the three high-resolution gamma spectrometry systems described in Section 2. The results are presented in Table 1. Inter-laboratory differences were evaluated using the maximum relative difference between laboratories for each sample. Relative differences for Ra-226 ranged from 3% to 30%, for Th-232 from 2% to 10%, and for K-40 from 1% to 5%, depending on material type and activity level. These variations are consistent with differences in detector characteristics, counting statistics, calibration procedures, and analytical uncertainties. The Gamma Index (I_γ) was calculated for each laboratory and is summarised in Table 2, including associated uncertainties. Although minor differences in Ra-226, Th-232, and K-40 activity concentrations were observed between laboratories, their impact on the resulting Gamma Index values was limited. Across all samples, I_γ ranged from 0.1 to 0.4, well below the European Commission

Table 1

Activity concentrations of Ra-226, Th-232, and K-40 (Bq·kg⁻¹) in first-phase building material samples measured by EPA, UCD, and UC laboratories.

Sample	Laboratory	Ra-226 (Bq·kg ⁻¹)	Th-232 (Bq·kg ⁻¹)	K-40 (Bq·kg ⁻¹)
1964 Concrete 1	EPA	17.1 ± 1.2	4.2 ± 1.0	30.5 ± 1.7
	UCD	18.1 ± 0.4	5.0 ± 0.6	34.0 ± 3.0
	UC	13.4 ± 1.6	3.3 ± 0.1	32.5 ± 6.9
1964 Concrete 2	EPA	5.5 ± 0.8	5.5 ± 0.9	53.7 ± 2.0
	UCD	7.1 ± 0.3	5.0 ± 0.6	53.0 ± 3.0
	UC	6.7 ± 1.2	5.0 ± 1.8	71.0 ± 9.0
1964 Concrete 3	EPA	32.9 ± 1.7	4.8 ± 1.3	66.0 ± 2.0
	UCD	34.0 ± 0.7	5.5 ± 0.7	71.0 ± 4.0
	UC	34.0 ± 3.0	4.1 ± 0.4	72 ± 11
1964 Plaster	EPA	13.0 ± 1.3	17.2 ± 1.8	670 ± 10
	UCD	13.7 ± 0.4	15.9 ± 0.9	635 ± 22
	UC	9.7 ± 1.6	16.0 ± 4.0	468 ± 45
Sand 1	EPA	12.1 ± 1.0	2.6 ± 0.8	57.4 ± 1.9
	UCD	14.1 ± 0.4	3.4 ± 0.5	65.0 ± 4.0
	UC	11.0 ± 1.4	3.3 ± 0.1	65.0 ± 9.0
Sand 2	EPA	14.1 ± 1.2	8.1 ± 1.3	465 ± 8
	UCD	14.8 ± 0.4	9.0 ± 0.7	453 ± 16
	UC	11.9 ± 1.5	8.9 ± 2.8	462 ± 43
Aggregates 1	EPA	29.3 ± 1.5	1.1 ± 0.1	7.8 ± 1.2
	UCD	30.6 ± 0.6	3.0 ± 0.1	5.4 ± 1.9
	UC	23.1 ± 2.3	3.2 ± 0.1	3.8 ± 0.1
Aggregates 2	EPA	31.9 ± 1.6	30.6 ± 0.6	6.3 ± 1.9
	UCD	30.1 ± 0.6	23.1 ± 2.3	5.1 ± 1.9
	UC	26.2 ± 2.5	2.6 ± 0.1	13.4 ± 0.1

Table 2

Gamma Index (I_γ) for first-phase building material samples measured by EPA, UCD, and UC laboratories.

Sample	Laboratory	Gamma Index (I _γ)
1964 Concrete 1	EPA	0.06 ± 0.01
	UCD	0.07 ± 0.01
	UC	0.07 ± 0.01
1964 Concrete 2	EPA	0.08 ± 0.01
	UCD	0.09 ± 0.01
	UC	0.07 ± 0.01
1964 Concrete 3	EPA	0.15 ± 0.01
	UCD	0.16 ± 0.01
	UC	0.15 ± 0.01
1964 Plaster	EPA	0.35 ± 0.01
	UCD	0.33 ± 0.01
	UC	0.27 ± 0.02
Sand 1	EPA	0.07 ± 0.01
	UCD	0.10 ± 0.01
	UC	0.09 ± 0.01
Sand 2	EPA	0.24 ± 0.01
	UCD	0.24 ± 0.01
	UC	0.23 ± 0.01
Aggregates 1	EPA	0.10 ± 0.01
	UCD	0.11 ± 0.01
	UC	0.10 ± 0.01
Aggregates 2	EPA	0.09 ± 0.01
	UCD	0.10 ± 0.01
	UC	0.10 ± 0.01

screening criterion of I_γ = 1, confirming that all materials are suitable for unrestricted use in construction.

3.1.1. Radionuclide activity concentrations, first phase

Concrete samples obtained from residential demolition sites dating back to 1964 showed Ra-226 activity concentrations ranging from 5.5 ± 0.8 Bq·kg⁻¹ (Concrete 2, EPA) to 34.0 ± 0.7 Bq·kg⁻¹ (Concrete 3, UCD). Corresponding Th-232 activity concentrations ranged from below detection limits (<3.30–4.10 Bq·kg⁻¹, UC) to 5.5 ± 0.7 Bq·kg⁻¹ (Concrete 3, UCD). K-40 activity concentrations in concrete varied between 30.5 ± 1.7 Bq·kg⁻¹ (Concrete 1, EPA) and 72 ± 11 Bq·kg⁻¹ (Concrete 3, UC).

Plaster samples exhibited comparatively elevated K-40 activity

concentrations, with values of 670 ± 10 Bq·kg⁻¹ (EPA) and 635 ± 22 Bq·kg⁻¹ (UCD), while a lower value of 468 ± 45 Bq·kg⁻¹ was reported by UC. Ra-226 activity concentrations in plaster ranged from 9.7 ± 1.6 Bq·kg⁻¹ to 13.7 ± 0.4 Bq·kg⁻¹, and Th-232 activities were consistently higher than those observed in concrete, varying between 15.9 ± 0.9 Bq·kg⁻¹ and 17.2 ± 1.8 Bq·kg⁻¹.

Sand samples displayed a wide range of K-40 activity concentrations depending on the source. Sand 1 exhibited K-40 activities between 57.4 ± 1.9 Bq·kg⁻¹ and 65 ± 9 Bq·kg⁻¹, whereas Sand 2 showed substantially higher K-40 activity, ranging from 453 ± 16 Bq·kg⁻¹ to 465 ± 8 Bq·kg⁻¹. Th-232 activity concentrations in sands ranged from 2.6 ± 0.8 Bq·kg⁻¹ (Sand 1, EPA) to 9.0 ± 0.7 Bq·kg⁻¹ (Sand 2, UCD), while Ra-226 activities varied between 11.0 ± 1.4 Bq·kg⁻¹ and 14.8 ± 0.4 Bq·kg⁻¹.

Aggregate samples showed Ra-226 activity concentrations between 26.2 ± 2.5 Bq·kg⁻¹ (UC) and 31.9 ± 1.6 Bq·kg⁻¹ (EPA). Th-232 activity concentrations were low, remaining below detection limits or close to 1 Bq·kg⁻¹, while K-40 activity concentrations ranged from 5.1 ± 1.9 Bq·kg⁻¹ to below 13.4 Bq·kg⁻¹.

3.1.2. Gamma Index, first phase

The Gamma Index (I_γ) values calculated for all samples in the first phase are presented in Table 2 and Fig. 1. Across all materials and participating laboratories, I_γ values ranged from 0.1 to 0.4, remaining well below the European Commission screening criterion of I_γ = 1.

Concrete samples exhibited I_γ values between 0.1 and 0.2, plaster samples between 0.3 and 0.4, sand samples between 0.1 and 0.3, and aggregate samples consistently showed values close to 0.1. These results demonstrate good inter-laboratory agreement and confirm that all materials analysed in this phase of the study are suitable for unrestricted use in construction from a radiological protection perspective.

3.2. Second phase

Activity concentrations of Ra-226, Th-232, and K-40 in the second-phase building material samples were measured independently by the EPA, UCD, and UC laboratories. This phase included concrete and one aggregate, both separated into multiple grain-size fractions to assess the effect of particle size. In addition, aggregates from low- and high-activity quarry sources, cementitious additives (GGBS and two fly ash samples), and imported ceramic tiles commonly used in Ireland. The measured activity concentrations and corresponding Gamma Index (I_γ) values are presented in Table 3 and Fig. 2.

3.2.1. Radionuclide activity concentrations, second phase

3.2.1.1. Grain-size effects. Particle size can influence gamma spectrometry through self-absorption and geometry effects, as well as radon loss affecting Ra-226, which can impact the Gamma Index (I_γ) (Jodlowski, 2006; Elnobi et al., 2017; Barba-Lobo, 2025). Concrete samples fractionated into <1.18 mm, 1.18–2 mm, and >2 mm showed minimal variation in activity concentrations (Ra-226: 27.5–29.4 Bq·kg⁻¹; Th-232: ≤5.7 Bq·kg⁻¹; K-40: 29–65 Bq·kg⁻¹; I_γ: 0.11–0.14), indicating that particle size has a negligible effect. Aggregates analysed at 1 mm and 1 cm showed larger Ra-226 differences (14.6–37.6 Bq·kg⁻¹), likely due to radon escape, while Th-232 and K-40 remained stable. Self-absorption corrections were applied, ensuring reliable determination of activity concentrations and Gamma Index values.

3.2.1.2. Influence of geological source on aggregates. Two additional aggregates, labelled Aggregate 1 and Aggregate 2, were sourced from quarries classified by Tellus data as low- and high-activity geological regions, respectively. Aggregate 1 contained Ra-226 18.6 Bq·kg⁻¹, Th-232 11.9 Bq·kg⁻¹, and K-40 219 Bq·kg⁻¹ (I_γ = 0.19), while Aggregate 2 was enriched in radionuclides with Ra-226 169 Bq·kg⁻¹, Th-232 206 Bq·kg⁻¹, and K-40 1155 Bq·kg⁻¹ (I_γ = 1.98), demonstrating the

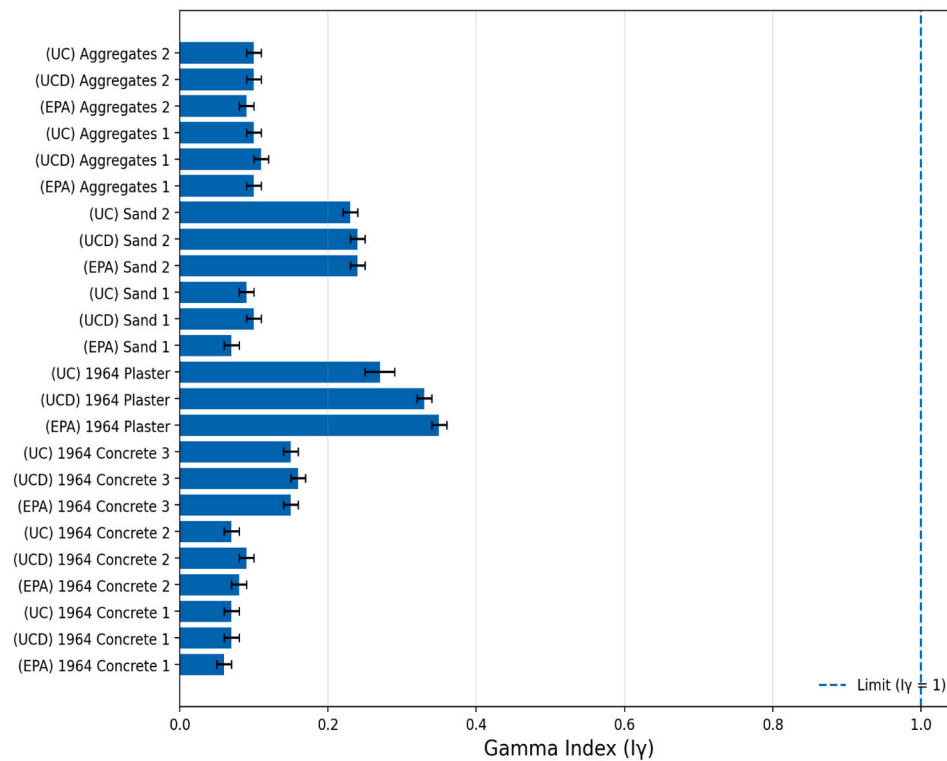


Fig. 1. Comparison of Gamma Index values measured by EPA, UCD, and UC for phase-1, building material samples.

Table 3

Activity concentrations of Ra-226, Th-232, and K-40 ($\text{Bq}\cdot\text{kg}^{-1}$) and corresponding Gamma Index (I_γ) for second-phase building material samples measured by EPA and UCD laboratories.

Sample	Ra-226 ($\text{Bq}\cdot\text{kg}^{-1}$)	Th-232 ($\text{Bq}\cdot\text{kg}^{-1}$)	K-40 ($\text{Bq}\cdot\text{kg}^{-1}$)	I_γ
Concrete (<1.18 mm)	28.1 ± 2.0	5.7 ± 1.2	64.9 ± 2.8	0.14 ± 0.01
Concrete (1.18–2 mm)	27.5 ± 2.1	2.0 ± 1.0	29.4 ± 2.5	0.11 ± 0.01
Concrete (>2 mm)	29.4 ± 2.0	4.4 ± 1.1	50.3 ± 2.6	0.14 ± 0.01
Aggregates (1 cm)	37.6 ± 3.8	<4.90	62.0 ± 11.0	0.17 ± 0.01
Aggregates (1 mm)	14.6 ± 1.4	<2.20	56.0 ± 6.0	0.08 ± 0.01
Aggregate 1	18.6 ± 0.5	11.9 ± 0.8	219 ± 9.0	0.19 ± 0.01
Aggregate 2	169 ± 3.0	206 ± 6.0	1155 ± 40	1.98 ± 0.03
GGBS	92 ± 5.0	102 ± 8.0	112 ± 4.0	0.85 ± 0.04
Fly ash A	55 ± 3.0	50 ± 4.0	452 ± 9.0	0.60 ± 0.02
Fly ash B	36 ± 4.0	22 ± 3.0	758 ± 15	0.48 ± 0.02
Tiles 1	68 ± 4.0	70 ± 5.0	286 ± 4.0	0.67 ± 0.10
Tiles 2	69 ± 1.0	59 ± 2.0	486 ± 17	0.69 ± 0.01

dominant influence of geological origin on natural radionuclide levels in construction materials.

3.2.1.3. Cementitious additives. Cementitious materials displayed moderate to elevated activity levels. GGBS contained Ra-226 $\sim 91.6 \text{ Bq}\cdot\text{kg}^{-1}$, Th-232 $\sim 101.8 \text{ Bq}\cdot\text{kg}^{-1}$, and K-40 $\sim 111.6 \text{ Bq}\cdot\text{kg}^{-1}$. Fly Ash A exhibited moderate Ra-226 and Th-232 activity, with K-40 $\sim 451.9 \text{ Bq}\cdot\text{kg}^{-1}$,

whereas Fly Ash B showed lower Ra-226 and Th-232 but very high K-40 ($\sim 758 \text{ Bq}\cdot\text{kg}^{-1}$). Beyond activity concentrations, the environmental performance of secondary cementitious materials is also influenced by their leaching behaviour. Previous studies on alkali-activated materials incorporating ground granulated blast furnace slag and phosphogypsum have shown that both radiological and non-radiological leaching generally remain low and within acceptable limits under realistic conditions, supporting their safe use in construction applications (Gijbels et al., 2019).

Notably, Fly Ash B contained detectable Cs-137 at $51.7 \pm 1.02 \text{ Bq}\cdot\text{kg}^{-1}$, slightly exceeding the $50 \text{ Bq}\cdot\text{kg}^{-1}$ threshold established by the Finnish Radiation and Nuclear Safety Authority (STUK, 2022). While this level is low from a radiological perspective, it suggests that additional exposure assessment may be warranted depending on the intended application.

Fly ash is widely recycled as a supplementary cementitious material in construction, particularly in concrete, mortars, and blended cements. Its incorporation can improve durability, reduce cement consumption, and provide a sustainable solution for industrial by-products. However, radionuclide content can vary depending on the source coal and combustion process, so monitoring for naturally occurring radionuclides and, in rare cases, artificial radionuclides (e.g., Cs-137) is important to ensure safety. Previous studies indicate that fly ash in construction typically remains well within international safety limits (Beretka and Mathew, 1985; Anagnostakis et al., 2004; UNSCEAR, 2000).

3.2.1.4. Imported ceramic tiles. Two imported ceramic tiles contained Ra-226 around $68\text{--}69 \text{ Bq}\cdot\text{kg}^{-1}$, Th-232 between 59 and $70 \text{ Bq}\cdot\text{kg}^{-1}$, and K-40 from 286 to $486 \text{ Bq}\cdot\text{kg}^{-1}$. These values are elevated relative to concrete and low-activity aggregates. As ceramic tiles are used as surface (superficial) materials rather than bulk materials, their I_γ values should be assessed against the reference level of $I_\gamma = 6$ specified in (European Commission, 1999) for surface applications, rather than the bulk criterion of $I_\gamma = 1$. Both tiles yield I_γ values well below 6 (0.67 ± 0.10 and 0.69 ± 0.01), confirming their radiological suitability as surface

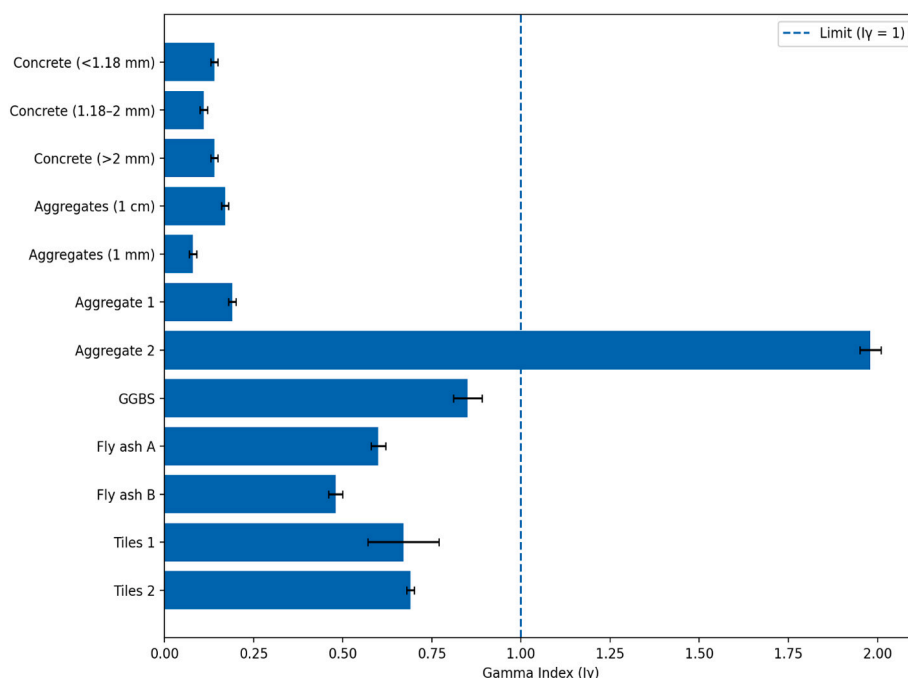


Fig. 2. Gamma Index (I_γ) of phase-2 building material samples. Note: the dashed line at $I_\gamma = 1$ is the reference criterion for bulk building materials. Ceramic tiles are surface (superficial) materials; their I_γ values (0.67 ± 0.10 and 0.69 ± 0.01) should be assessed against the higher reference level of $I_\gamma = 6$ specified for surface materials (European Commission, 1999), relative to which both tiles are well within acceptable limits.

materials.

3.2.2. Gamma Index, second phase

The Gamma Index (I_γ) was calculated for all second-phase building material samples (Table 3, Fig. 2) to assess potential radiological hazards. For materials subdivided by particle size, including concrete and one aggregate, I_γ values remained low (0.08–0.17), demonstrating that particle-size-related measurement effects have only a minor influence on the resulting Gamma Index and do not affect radiological classification.

Concrete fractions exhibited consistently low I_γ values (0.11–0.14), well below the European Commission screening limit of 1. The aggregate subdivided into two particle-size fractions showed slightly different I_γ values (0.08–0.17), primarily reflecting variations in measured Ra-226 activity due to radon escape from coarser particles, while Th-232 and K-40 contributions remained similar.

Aggregates sourced from different geological regions showed pronounced variability. Aggregate 1 yielded a low I_γ value of 0.19, whereas Aggregate 2 exhibited a substantially higher I_γ of 1.98, exceeding the EC screening criterion and highlighting the dominant influence of geological origin on radiological properties.

Cementitious additives displayed moderate I_γ values, with GGBS at 0.85 and Fly Ash A and B ranging from 0.48 to 0.60. Imported ceramic tiles exhibited slightly elevated I_γ values (0.67–0.69). All cementitious materials remained below the $I_\gamma = 1$ limit for unrestricted bulk use. Ceramic tiles, as surface materials, are appropriately assessed against the $I_\gamma = 6$ reference level (European Commission, 1999), relative to which their values are well within acceptable limits.

Overall, the Gamma Index results confirm that most materials investigated in the second phase comply with European radiological safety criteria, while high-activity aggregates require quarry-specific assessment prior to use in construction.

Comparison with the European database, which includes 57 samples from Ireland (Trevisi et al., 2012), shows that the activity concentrations measured in our 20 analysed materials, including conventional, demolition-derived, and recycled products fall within the reported ranges, with minor variations due to material type, particle size, and sample origin. The Gamma Index (I_γ) was calculated as a preliminary

assessment of radiological suitability, supporting the representativeness of the selected materials.

4. Conclusion

This study provides a comprehensive assessment of naturally occurring radioactive materials (NORM) in building materials commonly used in Ireland, addressing critical data gaps at both national and European levels. Across two experimental phases, 20 representative materials, including demolition-derived aggregates, construction sands, concretes, cementitious additives, recycled industrial by-products, and imported ceramic tiles, were analysed for Ra-226, Th-232, and K-40 activity concentrations using high-resolution gamma spectrometry, with measurements conducted in up to three independent laboratories.

Most conventional construction materials, including concretes, plaster, sands, and aggregates from low-activity geological regions, exhibited consistently low radionuclide concentrations and Gamma Index (I_γ) values well below the European Commission screening limit of 1. Particle-size fractionation experiments confirmed that grain size has minimal influence on the radiological classification of concrete and similar processed materials, supporting the robustness of standard measurement approaches and confirming their suitability for unrestricted use.

In contrast, aggregates from high-activity geological formations showed substantially elevated radionuclide concentrations (Ra-226 up to $169 \text{ Bq}\cdot\text{kg}^{-1}$, Th-232 up to $206 \text{ Bq}\cdot\text{kg}^{-1}$, K-40 up to $1155 \text{ Bq}\cdot\text{kg}^{-1}$), resulting in Gamma Index values approaching 2. This underscores the dominant influence of geological provenance on natural radioactivity levels and highlights the need for quarry-specific radiological screening before large-scale use or blending, in line with European Basic Safety Standards.

Cementitious additives and secondary materials, such as GGBS and fly ash, exhibited moderate but variable activity concentrations, remaining below regulatory limits for unrestricted use. The detection of trace Cs-137 slightly above the STUK reference level in one fly ash sample demonstrates the importance of targeted assessment for recycled and industrial by-products, whose radiological characteristics may

reflect fuel origin, combustion conditions, or legacy contamination.

Imported ceramic tiles showed elevated activity levels that, when assessed against the appropriate surface-material reference level of $I_\gamma = 6$ rather than the bulk criterion of $I_\gamma = 1$, are well within acceptable limits. This distinction highlights the importance of applying the correct reference value for surface materials and of periodic screening of imported manufactured products.

Inter-laboratory comparisons revealed high reproducibility, with differences consistent with expected analytical uncertainties, confirming the reliability of gamma spectrometry and supporting harmonised measurement protocols for national and European monitoring of NORM in construction materials.

From a European perspective, these findings are particularly relevant for sustainable construction, circular economy strategies, and the increasing reuse of demolition and industrial materials. While most materials pose minimal radiological concern, a graded, risk-based monitoring approach is recommended. Materials requiring enhanced investigation include high-activity aggregates, recycled construction and demolition materials, industrial by-products such as fly ash, and imported manufactured products. For these categories, batch-based screening, provenance documentation, and periodic verification can ensure compliance while enabling sustainable material reuse.

In conclusion, this work expands the national and European dataset on NORM in building materials, provides evidence-based guidance aligned with European regulatory frameworks, and supports the safe integration of low-carbon and recycled materials into the construction sector. The findings contribute to informed material selection, enhanced radiological protection, and the development of harmonised European guidelines that balance safety, sustainability, and environmental responsibility.

CRedit authorship contribution statement

R.E. Djabou: Writing – original draft. **M. Dodek:** Writing – review & editing. **C. Sainz:** Writing – review & editing, Funding acquisition, Formal analysis, Data curation, Conceptualization. **I. Fuente:** Writing – review & editing, Formal analysis, Data curation. **L. León Vintró:** Writing – review & editing, Methodology, Formal analysis, Data curation. **O. Hanley:** Writing – review & editing, Methodology, Formal analysis, Data curation. **J. Goggins:** Writing – review & editing, Resources, Project administration, Funding acquisition. **M. Foley:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) for language refinement. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Funding

This project is funded under the EPA Research Programme 2021–2030 (Grant No: 2023-GCE-1220). The EPA Research Programme is a Government of Ireland initiative funded by the Department of the Environment, Climate and Communications.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgments

The authors acknowledge the support of the Environmental Protection Agency (EPA), University College Dublin (UCD), and University of Cantabria (UC) laboratories for providing access to high-resolution gamma spectrometry facilities and technical guidance throughout the study.

Data availability

Data will be made available on request.

References

- Anagnostakis, M.J., Bolzan, C., De Felice, P., Fazio, A., Grisanti, G., Risica, S., Turtiainen, T., van der Graaf, E., 2004. A preliminary intercomparison of gamma-ray spectrometry on building materials. *Appl. Radiat. Isot.* 61, 381–388. <https://doi.org/10.1016/j.apradiso.2004.03.011>.
- Barba-Lobo, A., 2025. Self-attenuation effect correction methods employed for gamma-ray spectrometry: a critical review study. *Radiat. Phys. Chem.* 232, 112665. <https://doi.org/10.1016/j.radphyschem.2025.112665>.
- Beretka, J., Mathew, P.J., 1985. Natural radioactivity of Australian building materials, industrial wastes and by-products. *Health Phys.* 48, 87–95. <https://doi.org/10.1097/00004032-198501000-00007>.
- Canberra Industries, 2000. *Genie 2000 Spectroscopy Software, Version 3.2*. Canberra Industries, Meriden, CT, USA.
- Currie, L.A., 1968. Limits for qualitative detection and quantitative determination. *Anal. Chem.* 40 (3), 586–593. <https://doi.org/10.1021/ac60259a007>.
- Djabou, R.E., Mavon, C., Belafrites, A., Groetz, J.E., 2022. Mining treatment effects on natural radioactivity and radiological hazard index assessment in phosphates and fertilizers used in Algeria. *J. Radioanal. Nucl. Chem.* 331, 2081–2092. <https://doi.org/10.1007/s10967-022-08258-5>.
- Djabou, R.E., Dodek, M., Sainz, C., Fuente, I., León Vintró, L., Hanley, O., Goggins, J., Foley, M., 2026. Investigation of naturally-occurring radioactive materials (NORM) in building materials in Ireland. *Eur. Phys. J. Spec. Top.* <https://doi.org/10.1140/epjs/s11734-026-02353-4>.
- Elnohi, S., Harb, S., Ahmed, N.K., 2017. Influence of grain size on radionuclide activity concentrations and radiological hazard of building material samples. *Appl. Radiat. Isot.* 130, 43–48. <https://doi.org/10.1016/j.apradiso.2017.09.021>.
- El-Taher, A., 2010. Gamma spectroscopic analysis and associated radiation hazards of building materials used in Egypt. *Radiat. Prot. Dosim.* 138, 166–173. <https://doi.org/10.1093/rpd/ncp205>.
- European Commission, 1999. *Radiological protection principles concerning the natural radioactivity of building materials*. In: *Radiation Protection, Vol. 112*. Office for Official Publications of the European Communities, Luxembourg.
- European Commission, 2014. *Council Directive 2013/59/Euratom, Annex VIII — Basic Safety Standards for Protection Against the Dangers Arising from Exposure to Ionising Radiation*.
- European Committee for Standardisation (CEN), 2018. *CEN/TS 17216:2018 — Construction Products: Determination of Ra-226, Th-232, and K-40 Activity Concentrations Using Gamma-ray Spectrometry*. CEN, Brussels, Belgium.
- European Committee for Standardisation (CEN), 2022. *EN 17637:2022 — Construction Products: Assessment of Natural Radioactivity in Building Materials*.
- Fuente, M., Long, S., Fenton, D., Hung, L.C., Goggins, J., Foley, M., 2020. Review of recent radon research in Ireland, OPTI-SDS project and its impact on the National Radon Control Strategy. *Appl. Radiat. Isot.* 163, 109210. <https://doi.org/10.1016/j.apradiso.2020.109210>.
- Geological Survey Ireland, 2014. *Tellus Project: Airborne Geophysical and Geochemical Survey of Ireland*. Geological Survey Ireland, Dublin. <https://www.gsi.ie/en-ie/programmes-and-projects/tellus/Pages/default.aspx>.
- Gijbels, K., Landsberger, S., Samyn, P., Iacobescu, R.I., Pontikes, Y., Schreurs, S., Schroevers, W., 2019. Radiological and non-radiological leaching assessment of alkali-activated materials containing ground granulated blast furnace slag and phosphogypsum. *Sci. Total Environ.* 660, 1308–1321. <https://doi.org/10.1016/j.scitotenv.2019.01.089>.
- Gilmore, G., 2008. *Practical Gamma-ray Spectrometry*, 2nd ed. Wiley, Chichester.
- IAEA, 2010. *Measurement of Radionuclides in Food and the Environment: A Guidebook*. In: *Technical Reports Series, Vol. 295*. International Atomic Energy Agency, Vienna.
- ISO, 2015. *ISO 18589-2: Measurement of Radioactivity in the Environment — Soil — Part 2: Guidance for the Selection of the Sampling Strategy, Sampling and Pre-treatment of Samples*. International Organization for Standardization, Geneva.
- Jodłowski, P., 2006. Self-absorption correction in gamma-ray spectrometry of environmental samples — an overview. *Nukleonika* 51 (Suppl. 1), S21–S25. https://www.ichtj.waw.pl/ichtj/nukleon/back/full/vol51_2006/v51s2p21f.pdf.
- Karopoulos, K., Domingos, F., de With, G., Michalik, B., Okyar, H.B., Maramathas, C., Salpadimos, N., Potiriadis, C., Neculae, V., Durecová, A., Drábová, V., Pinak, M., 2025. Results of the joint IAEA/EEAE intercomparison exercise on radioanalytical characterization of NORM samples in the European region. *Radiat. Prot. Dosim.* 201, 223–246. <https://doi.org/10.1093/rpd/ncaf003>.
- Knoll, G.F., 2010. *Radiation Detection and Measurement*, 4th ed. Wiley, New York.

- Lee, E.M., Menezes, G., Finch, E.C., 2004. Natural radioactivity in building materials in the Republic of Ireland. *Health Phys.* 86, 378–383. <https://doi.org/10.1097/00004032-200404000-00007>.
- Madruga, M.J., Miró, C., Reis, M., Silva, L., 2019. Radiation exposure from natural radionuclides in building materials. *Radiat. Prot. Dosim.* 185, 49–57. <https://doi.org/10.1093/rpd/ncy256>.
- Markkanen, M., 1995. Radiation dose assessments for materials with elevated natural radioactivity. In: STUK-B-STO 32. Finnish Centre for Radiation and Nuclear Safety (STUK), Helsinki. <https://www.osti.gov/etdweb/servlets/purl/440009>.
- Michalik, B., Dvorzhak, A., Pereira, R., Lourenço, J., Haanes, H., Di Carlo, C., Nuccetelli, C., Venoso, G., Leonardi, F., Trevisi, R., Trotti, F., Ugolini, R., Pannecoucke, L., Blanchart, P., Perez-Sanchez, D., Real, A., Escibano, A., Fevrier, L., Kallio, A., Skipperud, L., et al., 2023. A methodology for the systematic identification of naturally occurring radioactive materials (NORM). *Sci. Total Environ.* 885, 163324. <https://doi.org/10.1016/j.scitotenv.2023.163324>.
- Mrdakovic Popic, J., Urso, L., Michalik, B., et al., 2023. Assessing the exposure situations with naturally occurring radioactive materials across European countries by means of the e-NORM survey. *Sci. Total Environ.* 905, 167065. <https://doi.org/10.1016/j.scitotenv.2023.167065>.
- ORTEC, 2010. *EFFTRAN: Self-attenuation Correction Software for Gamma Spectrometry*. ORTEC, Oak Ridge, TN, USA.
- Paepen, J., Van Ammel, R., Hult, M., Marissens, G., Stroh, H., Emteborg, H., Seghers, J., 2024. Results of the Collaborative Assessment Experiment for the Validation of CEN/TS 17216 Precision. Publications Office of the European Union. <https://doi.org/10.2760/3342>.
- Papaefthymiou, H., Gouseti, O., 2008. Natural radioactivity and associated radiation hazards in building materials used in Peloponnese, Greece. *Radiat. Meas.* 43, 1453–1457. <https://doi.org/10.1016/j.radmeas.2008.03.032>.
- Papastefanou, C., Stoulos, S., Manolopoulou, M., 2005. The radioactivity of building materials. *J. Radioanal. Nucl. Chem.* 266, 367–372. <https://doi.org/10.1007/s10967-005-0918-z>.
- Quindós, L., Sainz, C., Fuente, I., et al., 2006. Correction by self-attenuation in gamma-ray spectrometry for environmental samples. *J. Radioanal. Nucl. Chem.* 270, 339–343. <https://doi.org/10.1007/s10967-006-0354-8>.
- Righi, S., Bruzzi, L., 2006. Natural radioactivity and radon exhalation in building materials used in Italian dwellings. *J. Environ. Radioact.* 88, 158–170. <https://doi.org/10.1016/j.jenvrad.2006.01.009>.
- Rodrigues e Silva, C., Monteiro de Oliveira, F., 2023. Natural radioactivity in mineral phosphate fertilizers and its impacts on human health: an overview. *Environ. Sci. Pollut. Res.* 30, 118149–118160. <https://doi.org/10.1007/s11356-023-30467-y>.
- Sima, O., Arnold, D., Dovlete, C., 2001. GESPECOR: a versatile tool in gamma-ray spectrometry. *J. Radioanal. Nucl. Chem.* 248, 359–364. <https://doi.org/10.1023/A:1010619806898>.
- STUK, 2022. Radiation and Nuclear Safety Authority Regulation on Practices That Cause Exposure to Natural Radiation. Radiation and Nuclear Safety Authority, Finland. Available at: <https://www.stuklex.fi/en/maarays/stuk-s-6-2022>.
- Trevisi, R., Risica, S., D'Alessandro, M., Paradiso, D., Nuccetelli, C., 2012. Natural radioactivity in building materials in the European Union: a database and an estimate of radiological significance. *J. Environ. Radioact.* 105, 11–20. <https://doi.org/10.1016/j.jenvrad.2011.10.001>.
- Turhan, Ş., 2008. Assessment of the natural radioactivity and radiological hazards in Turkish cement and its raw materials. *J. Environ. Radioact.* 99, 404–414. <https://doi.org/10.1016/j.jenvrad.2007.11.001>.
- UNSCEAR, 2000. *Sources and Effects of Ionizing Radiation*. United Nations Scientific Committee on the Effects of Atomic Radiation, United Nations, New York.
- UNSCEAR, 2008. *Sources and Effects of Ionizing Radiation*, Updated Version. United Nations, New York.